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Pixel layouts for electronic device displays

Abstract

Increasing resolution of liquid crystal displays may result in small distances between adjacent liquid crystal display pixels. This tight pixel spacing may reduce transmission through the liquid crystal display pixels and may result in cross-talk between the liquid crystal display pixels. To increase transmission and, correspondingly, display efficiency, a reflective layer may be included in the liquid crystal display. The reflective layer recycles backlight that may otherwise be absorbed, improving transmittance and efficiency. To reduce color shift and color mixing caused by cross-talk, the pixels may have their pixel electrodes arranged in a zigzag layout. Each pixel electrode may have a height that is less than or equal to the total height of the pixel divided by two. The pixel electrodes in a given row are also alternately coupled to first and second gate lines. This zigzag layout results in an increased distance between adjacent pixel electrodes, mitigating pixel cross-talk.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7084849	12/2005	Noguchi et al.	N/A	N/A
7982219	12/2010	Kuo et al.	N/A	N/A
8243238	12/2011	Imayama et al.	N/A	N/A
9070334	12/2014	Chen et al.	N/A	N/A
9104079	12/2014	Uehara et al.	N/A	N/A
9217872	12/2014	Niioka et al.	N/A	N/A
10559270	12/2019	Chi	N/A	G09G 3/3614
10636847	12/2019	Yu et al.	N/A	N/A
2016/0320666	12/2015	Kim	N/A	G02F 1/136286
2018/0114495	12/2017	Jung	N/A	G09G 3/3607
2018/0120656	12/2017	Okada et al.	N/A	N/A
2020/0073167	12/2019	Shi et al.	N/A	N/A
2021/0364840	12/2020	Cao	N/A	N/A
2022/0317495	12/2021	Feng et al.	N/A	N/A

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Background/Summary

(1) This application is a continuation of non-provisional patent application Ser. No. 17/516,499, filed Nov. 1, 2021, which claims the benefit of provisional patent application No. 63/120,073, filed Dec. 1, 2020, which are hereby incorporated by reference herein in their entireties.

BACKGROUND

(1) This relates generally to electronic devices, and, more particularly, to electronic devices with displays.

(2) Electronic devices often include displays. For example, an electronic device may have a liquid

crystal display (LCD) based on liquid crystal display pixels. In this type of display, each pixel includes a pixel electrode that selectively applies an electric field to liquid crystal material. This selectively modifies the polarization of backlight that passes through the liquid crystal material, which allows for the display pixels to control the intensity of emitted light.

(3) To control a given pixel in a liquid crystal display, a voltage may be applied to the pixel electrode of the given pixel. Ideally, the voltage at the pixel electrode of the given pixel would not affect any neighboring pixels. However, there may be pixel cross-talk that allows nominally 'off' pixels to emit light due to an adjacent 'on' pixel's leakage. The pixel cross-talk may degrade display performance and cause a color-shift in the resulting image.

(4) It may be desirable to reduce the distance between pixels in a display in order to increase the resolution of the display. However, pixel cross-talk between adjacent LCD pixels may worsen as the distance between pixels decreases.

(5) It would therefore be desirable to be able to provide improved displays for electronic devices.

SUMMARY

(6) An electronic device may have a display such as a liquid crystal display. The liquid crystal display (LCD) may have an array of liquid crystal display pixels formed by a liquid crystal layer that is interposed between transparent substrates and controlled by pixel electrodes.

(7) Each liquid crystal display pixel may have a respective pixel electrode. Voltage may be applied to the pixel electrode of each liquid crystal display pixel to control how much backlight passes through each liquid crystal display pixel. Small distances between adjacent liquid crystal display pixels may reduce transmission through the liquid crystal display pixels and may result in cross-talk between the liquid crystal display pixels.

(8) To increase transmission and, correspondingly, display efficiency, a reflective layer may be included in the liquid crystal display. The reflective layer may have approximately the same footprint as an opaque masking layer that defines the light-emitting area of each pixel. The reflective layer recycles backlight that may otherwise be absorbed, improving transmittance and efficiency.

(9) To reduce color shift and color mixing caused by cross-talk, the pixels may have their pixel electrodes arranged in a zigzag layout. Each pixel electrode may have a height that is less than or equal to the total height of the pixel divided by two. The pixel electrodes in a given row are also alternately coupled to first and second adjacent gate lines. This zigzag layout results in an increased distance between adjacent pixel electrodes, mitigating pixel cross-talk.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of an illustrative electronic device having a display in accordance with an embodiment.

(2) FIG. 2 is a schematic diagram of an illustrative display in accordance with an embodiment.

(3) FIG. 3 is a schematic diagram of an illustrative display with liquid crystal display pixels in accordance with an embodiment.

(4) FIG. 4 is a cross-sectional side view of an illustrative display having a backlight and liquid crystal display layers in accordance with an embodiment.

(5) FIG. 5 is a cross-sectional side view of illustrative liquid crystal display layers that include a reflective metal layer underneath an opaque masking layer in accordance with an embodiment.

(6) FIG. 6 is a top view of an illustrative display having a zigzag layout of pixel electrodes in accordance with an embodiment.

DETAILED DESCRIPTION

(7) An illustrative electronic device of the type that may be provided with a display is shown in

FIG. 1. Electronic device **10** may be a computing device such as a laptop computer, a computer monitor containing an embedded computer, a tablet computer, a cellular telephone, a media player, or other handheld or portable electronic device, a smaller device such as a wrist-watch device, a pendant device, a headphone or earpiece device, a device embedded in eyeglasses or other equipment worn on a user's head, or other wearable or miniature device, a display, a computer display that contains an embedded computer, a computer display that does not contain an embedded computer, a gaming device, a navigation device, an embedded system such as a system in which electronic equipment with a display is mounted in a kiosk or automobile, or other electronic equipment. Electronic device **10** may have the shape of a pair of eyeglasses (e.g., supporting frames), may form a housing having a helmet shape, or may have other configurations to help in mounting and securing the components of one or more displays on the head or near the eye of a user. As examples, electronic device **10** may be an augmented reality (AR) headset and/or virtual reality (VR) headset.

(8) As shown in **FIG. 1**, electronic device **10** may include control circuitry **16** for supporting the operation of device **10**. The control circuitry may include storage such as hard disk drive storage, nonvolatile memory (e.g., flash memory or other electrically-programmable-read-only memory configured to form a solid state drive), volatile memory (e.g., static or dynamic random-access memory), etc. Processing circuitry in control circuitry **16** may be used to control the operation of device **10**. The processing circuitry may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors, power management units, audio chips, application specific integrated circuits, etc.

(9) Input-output circuitry in device **10** such as input-output devices **12** may be used to allow data to be supplied to device **10** and to allow data to be provided from device **10** to external devices. Input-output devices **12** may include buttons, joysticks, scrolling wheels, touch pads, key pads, keyboards, microphones, speakers, tone generators, vibrators, cameras, sensors, light-emitting diodes and other status indicators, data ports, etc. A user can control the operation of device **10** by supplying commands through input-output devices **12** and may receive status information and other output from device **10** using the output resources of input-output devices **12**.

(10) Input-output devices **12** may include one or more displays such as display **14**. Display **14** may be a touch screen display that includes a touch sensor for gathering touch input from a user or display **14** may be insensitive to touch. A touch sensor for display **14** may be based on an array of capacitive touch sensor electrodes, acoustic touch sensor structures, resistive touch components, force-based touch sensor structures, a light-based touch sensor, or other suitable touch sensor arrangements. A touch sensor for display **14** may be formed from electrodes formed on a common display substrate with the pixels of display **14** or may be formed from a separate touch sensor panel that overlaps the pixels of display **14**. If desired, display **14** may be insensitive to touch (i.e., the touch sensor may be omitted). Display **14** in electronic device **10** may be a head-up display that can be viewed without requiring users to look away from a typical viewpoint or may be a head-mounted display that is incorporated into a device that is worn on a user's head. If desired, display **14** may also be a holographic display used to display holograms.

(11) Control circuitry **16** may be used to run software on device **10** such as operating system code and applications. During operation of device **10**, the software running on control circuitry **16** may display images on display **14**.

(12) **FIG. 2** is a diagram of an illustrative display. As shown in **FIG. 2**, display **14** may include layers such as substrate layer **26**. Substrate layers such as layer **26** may be formed from rectangular planar layers of material or layers of material with other shapes (e.g., circular shapes or other shapes with one or more curved and/or straight edges). The substrate layers of display **14** may include glass layers, polymer layers, silicon layers, composite films that include polymer and inorganic materials, metallic foils, etc.

(13) Display **14** may have an array of pixels **22** for displaying images for a user such as pixel array

28. Pixels 22 in array 28 may be arranged in rows and columns. The edges of array 28 may be straight or curved (i.e., each row of pixels 22 and/or each column of pixels 22 in array 28 may have the same length or may have a different length). There may be any suitable number of rows and columns in array 28 (e.g., ten or more, one hundred or more, or one thousand or more, etc.). Display 14 may include pixels 22 of different colors. As an example, display 14 may include red pixels, green pixels, and blue pixels.

(14) Display driver circuitry 20 may be used to control the operation of pixels 28. Display driver circuitry 20 may be formed from integrated circuits, thin-film transistor circuits, and/or other suitable circuitry. Illustrative display driver circuitry 20 of FIG. 2 includes display driver circuitry 20A and additional display driver circuitry such as gate driver circuitry 20B. Gate driver circuitry 20B may be formed along one or more edges of display 14. For example, gate driver circuitry 20B may be arranged along the left and right sides of display 14 as shown in FIG. 2.

(15) As shown in FIG. 2, display driver circuitry 20A (e.g., one or more display driver integrated circuits, thin-film transistor circuitry, etc.) may contain communications circuitry for communicating with system control circuitry over signal path 24. Path 24 may be formed from traces on a flexible printed circuit or other cable. The control circuitry may be located on one or more printed circuits in electronic device 10. During operation, control circuitry (e.g., control circuitry 16 of FIG. 1) may supply circuitry such as a display driver integrated circuit in circuitry with image data for images to be displayed on display 14. Display driver circuitry 20A of FIG. 2 is located at the top of display 14. This is merely illustrative. Display driver circuitry may be located at both the top and bottom of display 14 or in other portions of device 10.

(16) To display the images on pixels 22, display driver circuitry 20A may supply corresponding image data to data lines D while issuing control signals to supporting display driver circuitry such as gate driver circuitry 20B over signal paths 30. With the illustrative arrangement of FIG. 2, data lines D run vertically through display 14 and are associated with respective columns of pixels 22.

(17) Gate driver circuitry 20B (sometimes referred to as gate line driver circuitry or horizontal control signal circuitry) may be implemented using one or more integrated circuits and/or may be implemented using thin-film transistor circuitry on substrate 26. Horizontal control lines G (sometimes referred to as gate lines, scan lines, emission control lines, etc.) run horizontally through display 14. Each gate line G is associated with a respective row of pixels 22. If desired, there may be multiple horizontal control lines such as gate lines G associated with each row of pixels. Individually controlled and/or global signal paths in display 14 may also be used to distribute other signals (e.g., power supply signals, etc.).

(18) Gate driver circuitry 20B may assert control signals on the gate lines G in display 14. For example, gate driver circuitry 20B may receive clock signals and other control signals from circuitry 20A on paths 30 and may, in response to the received signals, assert a gate line signal on gate lines G in sequence, starting with the gate line signal G in the first row of pixels 22 in array 28. As each gate line is asserted, data from data lines D may be loaded into a corresponding row of pixels. In this way, control circuitry such as display driver circuitry 20A and 20B may provide pixels 22 with signals that direct pixels 22 to display a desired image on display 14. Each pixel 22 may have a light-emitting diode and circuitry (e.g., thin-film circuitry on substrate 26) that responds to the control and data signals from display driver circuitry 20.

(19) Gate driver circuitry 20B may include blocks of gate driver circuitry such as gate driver row blocks. Each gate driver row block may include circuitry such output buffers and other output driver circuitry, register circuits (e.g., registers that can be chained together to form a shift register), and signal lines, power lines, and other interconnects. Each gate driver row block may supply one or more gate signals to one or more respective gate lines in a corresponding row of the pixels of the array of pixels in the active area of display 14.

(20) Display 14 for device 10 may be a liquid crystal display, an organic light-emitting diode display, an electrophoretic display, a plasma display, an electrowetting display, a display formed

using other display technologies, or a display that uses two or more of these display technologies in a hybrid configuration.

(21) FIG. 3 shows an example of a liquid crystal display. As shown in FIG. 3, liquid crystal display pixels 22 may be arranged in an array having rows and columns. The circuitry of the pixel array (i.e., the rows and columns of pixel circuits for pixels 22) may be controlled using signals such as data line signals on data lines D and gate line signals on gate lines G.

(22) Pixels 22 may contain thin-film transistor circuitry. For example, pixels 22 may contain silicon thin-film transistor circuitry such as polysilicon transistor circuitry or amorphous silicon transistor circuitry, semiconducting oxide thin-film transistor circuitry such as indium gallium zinc oxide transistor circuitry, or other silicon or semiconducting-oxide transistor circuitry. Pixels 22 may also include associated electrode structures for producing electric fields across a liquid crystal layer in display 14. Each of pixels 22 may have one or more thin-film transistors. For example, each pixel 22 may have a respective thin-film transistor such as thin-film transistor 94 to control the application of electric fields to a respective pixel-sized portion 54 of a liquid crystal layer in display 14. Display 14 may contain a color filter layer having an array of color filter elements associated with respective pixels 22 and a thin-film transistor layer on which circuitry such as the circuitry of FIG. 3 is formed. A liquid crystal layer may be interposed between the color filter layer and the thin-film transistor layer. Other configurations for display 14 may be used, if desired. The use of a liquid crystal display technology for forming display 14 is merely illustrative.

(23) The thin-film transistor structures that are used in forming pixels 22 may be located on a thin-film transistor substrate such as a layer of glass. The thin-film transistor substrate and the structures of pixels 22 that are formed on the surface of the thin-film transistor substrate may collectively form a thin-film transistor layer in display 14.

(24) Gate driver circuitry may be used to generate gate signals on gate lines G. The gate driver circuitry may be formed from thin-film transistors on the thin-film transistor layer or may be implemented in separate integrated circuits. The data line signals on data lines D in display 14 carry analog image data (e.g., voltages with magnitudes representing pixel brightness levels). During the process of displaying images on display 14, a display driver integrated circuit may receive digital data from control circuitry in device 10 and may produce corresponding analog data signals. The analog data signals may be demultiplexed and provided to data lines D.

(25) The data line signals on data lines D are distributed to the columns of pixels 22. Gate line signals on gate lines G are provided to the rows of pixels 22 by associated gate driver circuitry.

(26) The circuitry of display 14 such as demultiplexer circuitry, gate driver circuitry, and the circuitry of pixels 22 may be formed from conductive structures (e.g., metal lines and/or structures formed from transparent conductive materials such as indium tin oxide) and may include transistors such as transistor 94 that are fabricated on the thin-film transistor substrate layer of display 14. The thin-film transistors may be, for example, silicon thin-film transistors or semiconducting-oxide thin-film transistors.

(27) One of pixels 22 may be located at the intersection of each gate line G and data line D in display 14. A data signal on each data line D may be supplied to terminal 96 from one of data lines D. Thin-film transistor 94 (e.g., a thin-film polysilicon transistor or an amorphous silicon transistor) may have a gate terminal such as gate 98 that receives gate line control signals on gate line signal path G. When a gate line control signal is asserted, transistor 94 will be turned on and the data signal at terminal 96 will be passed to node 100 as voltage V_p . Data for display 14 may be displayed in frames. Following assertion of the gate line signal in each row to pass data signals to the pixels of that row, the gate line signal may be deasserted. In a subsequent display frame, the gate line signal for each row may again be asserted to turn on transistor 94 and capture new values of V_p .

(28) Each pixel 22 may have a signal storage element such as capacitor 102 or other charge storage elements. Storage capacitor 102 may be used to store signal V_p in each pixel 22 between frames

(i.e., in the period of time between the assertion of successive gate signals).

(29) Display **14** may have a common electrode coupled to node **104**. The common electrode (which is sometimes referred to as the Vcom electrode) may be used to distribute a common electrode voltage such as common electrode voltage Vcom to nodes such as node **104** in pixels **22**. The Vcom electrode **104** may be implemented using a blanket film of a transparent conductive material such as indium tin oxide and/or a layer of metal that is sufficiently thin to be transparent.

(30) In each pixel **22**, capacitor **102** may be coupled between nodes **100** and **104**. A parallel capacitance (sometimes referred to as capacitance CO arises across nodes **100** and **104** due to electrode structures in pixel **22** that are used in controlling the electric field through the liquid crystal material **54** of the pixel. As shown in FIG. 3, electrode structures **106** (e.g., a display pixel electrode with multiple fingers or other display pixel electrode for applying electric fields to liquid crystal material **54**) may be coupled to node **100** (or a display pixel electrode may be formed at node **104**). The capacitance C.sub.LC across liquid crystal material **54** is associated with the capacitance between electrode structures **106** and common electrode Vcom at node **104**. During operation, electrode structures **106** may be used to apply a controlled electric field (i.e., a field having a magnitude proportional to $V_p - V_{com}$) across pixel-sized liquid crystal material **54** in pixel **22**. Due to the presence of storage capacitor **102** and the capacitance C.sub.LC of material **54**, the value of V_p (and therefore the associated electric field across liquid crystal material **54**) may be maintained across nodes **106** and **104** for the duration of the frame.

(31) The electric field that is produced across liquid crystal material **54** causes a change in the orientations of the liquid crystals in liquid crystal material **54**. This changes the polarization of light passing through liquid crystal material **54**. The change in polarization may, in conjunction with upper and lower polarizers in display **14**, be used in controlling the amount of light that is transmitted through each pixel **22** in display **14** (e.g., how much light from a backlight unit is transmitted through each pixel **22**).

(32) A cross-sectional side view of an illustrative configuration for display **14** of device **10** is shown in FIG. 4. As shown in FIG. 4, display **14** may include a backlight unit such as backlight unit **42** (sometimes referred to as a backlight or backlight structures) for producing backlight **44**. During operation, backlight **44** travels outwards (vertically upwards in dimension Z in the orientation of FIG. 4) and passes through pixel structures in display layers **46**. This illuminates any images that are being produced by the pixels for viewing by a user. For example, backlight **44** may illuminate images on display layers **46** that are being viewed by viewer **48** in direction **50**.

(33) In a configuration in which display layers **46** (sometimes referred to as display panel **46**) are used in forming a liquid crystal display, display layers **46** may include liquid crystal display layers **58** (sometimes referred to as display layers **58**) that are sandwiched between lower polarizer layer **52** and upper polarizer layer **56**. Liquid crystal display layers **58** may include a liquid crystal layer that is interposed between first and second substrates, as will be discussed in greater detail in connection with FIG. 5.

(34) Any desired arrangement may be used for backlight structures **42**. In one possible arrangement, the backlight structures include a light guide layer (formed from a transparent material such as clear glass or plastic). During operation of the backlight structures, a light source (e.g., one or more light-emitting diodes) may emit light into an edge of the light guide layer. The light is distributed throughout the light guide layer and is scattered upwards (in the positive Z-direction) to serve as backlight for the display. This type of arrangement may be referred to as an edge-lit backlight. Alternatively, a direct-lit backlight may be used. In a direct-lit backlight, an array of light-emitting diodes may be formed underneath the display layers. Each light-emitting diode may provide backlight for a respective portion (cell) of the display. One or more light spreading layers may be incorporated to avoid hotspots caused by the light-emitting diodes.

(35) Lower polarization layer **52** ensures that all backlight received by the liquid crystal display layers **58** have a uniform polarization. The liquid crystal display layers **58** may then selectively

modify the polarization of the light to control how much light passes through upper polarization layer **56**.

(36) To control a given pixel in a liquid crystal display, a voltage may be applied to the pixel electrode of the given pixel. Ideally, the voltage at the pixel electrode of the given pixel would not affect any neighboring pixels. However, there may be pixel cross-talk that allows nominally 'off' pixels to emit light due to an adjacent 'on' pixel's leakage. The pixel cross-talk may degrade display performance and cause a color-shift in the resulting image. Cross-talk may be exacerbated as spacing between adjacent pixels decreases (e.g., to provide high resolution displays). FIG. 5 is a cross-sectional side view of illustrative liquid crystal display layers that are arranged to mitigate cross-talk (even with tightly spaced pixels).

(37) As shown in FIG. 5, liquid crystal display layers **58** include an upper substrate **112** and a lower substrate **110**. Layers **110** and **112** may be formed from transparent substrate layers such as clear layers of glass, plastic, or any other desired material. Liquid crystal layer **54** may be sandwiched between substrates **110** and **112**.

(38) Pixel electrodes **106** control the electric field through the liquid crystal material **54**. The electric field that is produced across liquid crystal material **54** causes a change in the orientations of the liquid crystals in liquid crystal material **54**. This changes the polarization of light passing through liquid crystal material **54**. The change in polarization may, in conjunction with upper polarizer **56** and lower polarizer **52** (as shown in FIG. 4), be used in controlling the amount of light that is transmitted through each pixel **22** in display **14** (e.g., how much light from a backlight unit is transmitted through each pixel **22**). The liquid crystal display layers also include common electrode **104**. Common electrode **104** and pixel electrodes **106** are separated by passivation layer **122**. Passivation layer **122** (sometimes referred to as insulating layer **122**) may be formed from any desired material.

(39) Thin-film transistor circuitry **128** may be formed on substrate **110** and may be used to control the pixel electrodes **106**. As shown in FIG. 5, thin-film transistor circuitry **128** may include various conductive components such as signal lines **118** in FIG. 5 (e.g., gate lines and/or data lines as shown in FIGS. 2 and 3), thin-film transistors such as transistors **94** in FIG. 3, and any other desired components. One or more dielectric layers **130** may be incorporated to isolate the components of thin-film transistor layer **128** as desired.

(40) Color filter layer **114** may include a plurality of color filter elements **114-E**. Each color filter element is overlapping with a respective pixel **22**. Red color filter elements, green color filter elements, blue color filter elements, and/or color filter elements of other desired colors may be included in color filter layer **114**. An opaque masking layer **126** is formed between each adjacent pair of color filter elements. Opaque masking layer **126** (sometimes referred to as black masking layer **126**, black matrix **126**, etc.) prevents cross-talk between adjacent pixels in the display. One or more dielectric layers **120** (which may be formed from any desired material) may be interposed between color filter layer **114** (and opaque masking layer **126**) and common electrode **104**. An overcoat layer **124** (which may be formed from any desired material) may be interposed between the upper substrate **112** and liquid crystal layer **54**.

(41) The illustrative arrangement for pixel electrodes **106** and common electrode **104** are merely illustrative. In general, the liquid crystal display pixels **22** may operate using fringe-field switching, in-plane switching, or any other desired type of switching technique.

(42) The pixels of the liquid crystal display panel of FIG. 5 may be tightly spaced to enable a high resolution display. As shown, adjacent pixel electrodes **106** may be separated by a distance **134**. Distance **134** may be less than 20 microns, less than 10 microns, less than 6 microns, less than 5 microns, between 3 and 5 microns, between 2 and 10 microns, greater than 2 microns, greater than 4 microns, etc.

(43) Forming tightly spaced pixels of this type may reduce the transmission of backlight through the liquid crystal display panel. The reduced transmission may occur as a result of the opaque

masking layer **126** occupying a large part of each overall pixel area. In one example, the light-emitting area may occupy a smaller percentage of each pixel than the opaque masking layer. Having so much of each pixel occupied by the opaque masking layer decreases transmittance through the pixel and, accordingly, decreases efficiency of the display.

(44) Positioning color filter layer **114** between liquid crystal layer **54** and thin-film transistor layer **128** (as in FIG. 5) may increase efficiency of the display (even when the display has tightly spaced pixels). In the arrangement of FIG. 5, color filter layer **114** is also positioned between liquid crystal layer **54** and the backlight unit **42**. In another possible arrangement, the color filter layer **114** may be positioned between liquid crystal layer **54** and upper substrate **112**. However, instead using the arrangement of FIG. 5 may increase the transmittance through the pixels and the efficiency of the display. This arrangement of the color filter layer on the thin-film transistor layer may sometimes be referred to as a color filter on array (COA) LCD.

(45) Additional transmittance (and efficiency) improvements may be obtained by including reflective layer **116** in display **14**. As shown in FIG. 5, reflective layer **116** may be formed underneath opaque masking layer **126**. Reflective layer **116** may be formed from a metal (such as silver or aluminum) or any other desired material. Reflective layer **116** may have a reflectivity of greater than 80%, greater than 90%, greater than 95%, greater than 99%, etc.

(46) Reflective layer **116** may reflect light from backlight unit **42** that is subsequently recycled and passes through the light-emitting area of each pixel. Without reflective layer **116**, some of this light would instead be absorbed (e.g., by opaque masking layer **126**). Including reflective layer **116** increases the likelihood of backlight **44** (from backlight unit **42**) ultimately passing through the pixel aperture (due to the recycling caused by reflective layer **116**).

(47) In one possible arrangement, reflective layer **116** may have the same footprint as opaque masking layer **126**. Opaque masking layer **126** may have a width **136** whereas reflective layer **116** has a width **138**. Widths **136** and **138** of the layers between adjacent pixels may be the same or similar (e.g., within 10%, within 5%, within 1%, etc.). Opaque masking layer **126** directly overlaps reflective layer **116** in the vertical direction (e.g., parallel to the Z-direction). One or more dielectric layers **132** may be interposed between lower substrate **110** and thin-film transistor layer **128**.

(48) Having a small distance between adjacent pixel electrodes may lead to cross-talk between adjacent pixels. For example, consider the example where pixel **22-2** is intended to be 'on' (e.g., pass backlight with a maximum brightness) and pixels **22-1** and **22-3** are intended to be 'off' (e.g., pass no backlight). Applying voltage to the pixel electrode for **22-2** may cause pixel **22-2** to emit light (as desired). However, leakage current between pixel **22-2** and the adjacent pixels **22-1** and **22-3** may cause pixels **22-1** and **22-3** to also emit light (even though they are nominally off). This pixel cross-talk may have undesired effects such as color shift at off-axis viewing angles and color mixing at on-axis viewing angles.

(49) To mitigate cross-talk caused by tight spacing of the liquid crystal display pixels, a zigzag pixel arrangement may be used. FIG. 6 is a top view of an illustrative display showing how a zigzag pixel layout may be used to mitigate cross-talk. Columns and rows of pixels are arranged in the display of FIG. 6. The total pixel height and width may be uniform across the display. Each pixel includes an aperture **140** (sometimes referred to as opening **140** or light-emitting area **140**) that is surrounded by opaque masking layer **126**. Pixel apertures corresponding to red pixels are labeled R, pixel apertures corresponding to green pixels are labeled G, and pixel apertures corresponding to blue pixels are labeled B. Each pixel also includes a respective transistor **94** that is obscured by the opaque masking layer **126**. Similar to as shown in FIG. 3, each transistor **94** has a gate terminal **98** that is coupled to a respective gate line and a terminal **96** that is coupled to a respective data line. Transistor **94** may apply a desired voltage to a pixel electrode **106** that is overlapped by pixel aperture **140**.

(50) Cross-talk between adjacent pixels may be reduced by increasing the distance between adjacent pixel electrodes. In the zigzag arrangement of FIG. 6, adjacent pixel apertures **140** (and

corresponding pixel electrodes **106**) are separated by a first distance **142** in the X-direction as well as a second distance **144** in the Y-direction. The total distance **146** between adjacent pixel electrodes (e.g., the corner-to-corner distance) is therefore increased relative to an example where there is no vertical separation **144**.

(51) Corner-to-corner distance **146** may be greater than 2 microns, greater than 4 microns, greater than 6 microns, greater than 8 microns, less than 10 microns, between 2 and 10 microns, between 2 and 25 microns, etc.

(52) Said another way, each pixel electrode does not overlap the adjacent pixel electrode in the horizontal direction (e.g., parallel to the X-axis). Consider the pixel aperture in row N and column N+1. The pixel aperture for this green pixel has a lower edge that extends parallel to axis **150**. As shown, axis **150** is separated (in the Z-direction by distance **144**) from the axis **152** of the upper edge of the pixels in row N, column N and row N, column N+2. Axis **150** intersects (e.g., is colinear with) the lower edge of every other pixel electrode in its row.

(53) Axis **150** does not intersect the pixel electrode of every other pixel in its row.

(54) Each pixel electrode **106** is positioned between adjacent pixel transistors **94** (not between adjacent pixel electrodes **106**). This increases the separation between pixel electrodes to corner-to-corner distance **146**, which mitigates cross-talk between the pixels.

(55) A zigzag gate driving scheme is used to handle driving the pixels of FIG. **6**. As shown, each gate line is coupled to every other pixel in one row and every other pixel in another, adjacent row. Consider gate line N+1. This gate line is coupled to the red pixel in Row N+1, column N, the green pixel in row N, column N+1, the blue pixel in row N+1, column N+2, the red pixel in row N, column N+3, the green pixel in row N+1, column N+4, and the blue pixel in row N, column N+5. Instead of being coupled to every pixel in row N+1, gate line N+1 is coupled to every other pixel in row N+1 and every other pixel in row N. This pattern may be repeated for each gate line in the array. For each pixel electrode coupled to a respective gate line, a transistor (**94**) is interposed between that pixel electrode and the respective gate line.

(56) Each pixel may have a total height **154** (e.g., the dimension parallel to the Y-axis in FIG. **6**). Each pixel aperture (and pixel electrode) may also have a respective height **156** (e.g., the dimension parallel to the Y-axis in FIG. **6**). Height **156** may be less than or equal to height **154** divided by two ($H_{\text{sub.APERTURE}} \leq (H_{\text{sub.TOTAL}}/2)$). This ensures that pixel apertures in adjacent columns do not overlap in the X-direction (which increases the distance between the pixel electrodes to mitigate cross-talk).

(57) This property may also be described relative to the distance **158** in the Y-direction. As shown in FIG. **6**, distance **158** is a measure of the distance between pixel electrodes in adjacent rows. In other words, the opaque masking layer that is interposed between the lower edge of the pixel aperture in row N, column N+2 and the pixel aperture in row N+1, column N+2 has a dimension (width) **158**. Distance **158** may be greater than or equal to the height **156** of each pixel aperture. Again, this ensures that pixel apertures in adjacent columns do not overlap in the X-direction (which increases the distance between the pixel electrodes to mitigate cross-talk). With the properties described above, the edge of a first pixel aperture may, at closest, be colinear with the edge of a pixel aperture in an adjacent column.

(58) The pixel layout of FIG. **6** increases the distance between each adjacent pixel electrode in the display. Accordingly, cross-talk between the pixels is mitigated and color mixing and color shift is improved.

(59) It should be understood that display **14** in FIG. **6** has a cross-sectional arrangement as shown in FIG. **5**. The color filter layer in display **14** in FIG. **6** is interposed between the liquid crystal layer and the lower substrate (as in FIG. **5**). Additionally, opaque masking layer **126** in FIG. **6** overlaps a reflective layer (e.g., reflective layer **116** in FIG. **5**). The reflective layer may have the same footprint as the opaque masking layer **126** in FIG. **6**.

(60) These examples are merely illustrative. The pixel layout shown in FIG. **6** may instead be

applied to a display having any desired cross-sectional arrangement.

(61) The foregoing is merely illustrative and various modifications can be made by those skilled in the art without departing from the scope and spirit of the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. A display comprising: an array of pixels in rows and columns; a plurality of data lines that is coupled to the array of pixels, wherein each data line is coupled to a respective column of pixels; and a plurality of gate lines that is coupled to the array of pixels, wherein a first gate line of the plurality of gate lines is coupled to pixels in a first row, wherein a second gate line of the plurality of gate lines is coupled to alternating pixels in the first row and a second adjacent row, wherein the first gate line is adjacent to the second gate line, wherein the first gate line is coupled to a first pixel of a color in the first row and a first column, and wherein the second gate line is coupled to a second pixel of the color in the second row and the first column.
2. The display defined in claim 1 wherein the second gate line is coupled to a third pixel in the first row, a fourth pixel in the second row, and a fifth pixel in the first row.
3. The display defined in claim 2 wherein the third pixel is in a second column adjacent to the first column, the fourth pixel is in a third column adjacent to the second column, and the fifth pixel is in a fourth column adjacent to the third column.
4. The display defined in claim 1 wherein the second gate line is coupled to odd numbered columns in the first row and even numbered columns in the second, adjacent row.
5. The display defined in claim 1 wherein each pixel comprises a respective pixel electrode and wherein the pixel electrodes are arranged in a zigzag layout.
6. The display defined in claim 5, further comprising: an opaque masking layer; and a plurality of pixel apertures in the opaque masking layer that each overlap a respective pixel electrode, wherein the plurality of pixel apertures is arranged in the zigzag layout.
7. The display defined in claim 1 wherein each pixel comprises a pixel electrode and a transistor that is coupled between the respective pixel electrode and one of the plurality of gate lines.
8. The display defined in claim 7 wherein, for each pixel, the transistor is interposed between the pixel electrode and the one of the plurality of gate lines.
9. The display defined in claim 8, further comprising: an opaque masking layer; and a plurality of pixel apertures in the opaque masking layer that each overlap a respective pixel electrode, wherein each one of the plurality of gate lines extends in a first direction, wherein each pixel aperture has a first height that extends in a second direction that is orthogonal to the first direction, wherein the opaque masking layer between adjacent pixel apertures in the same column and different rows has a second height that extends in the second direction, and wherein the second height is greater than or equal to the first height.
10. The display defined in claim 7, further comprising: an opaque masking layer, wherein each one of the transistors is overlapped by the opaque masking layer.
11. The display defined in claim 8 wherein each one of the plurality of gate lines extends in a first direction, wherein each pixel has a first height that extends in a second direction that is orthogonal to the first direction, wherein each pixel electrode has a second height that extends in the second direction, and wherein the second height is less than or equal to the first height divided by two.
12. A display comprising: an array of pixels in rows and columns; a plurality of data lines that is coupled to the array of pixels, wherein each data line is coupled to each pixel in a respective column of pixels; a plurality of gate lines that is coupled to the array of pixels, wherein a given gate line of the plurality of gate lines is coupled to alternating pixels in first and second adjacent rows, wherein the given gate line is coupled to a first pixel of a first color in the first row and a first column, wherein the given gate line is coupled to a second pixel of a second color in the second

row and a second column, and wherein the second column is adjacent to the first column.

13. The display defined in claim 12 wherein the given gate line is coupled to a third pixel of the first color in the second row and a third column.

14. The display defined in claim 12 wherein each pixel comprises a respective pixel electrode, wherein the given gate line extends in a direction, and wherein adjacent pixel electrodes are non-overlapping in the direction.

15. A display comprising: an array of pixels in rows and columns, wherein each pixel comprises a pixel electrode and a transistor; a plurality of data lines that is coupled to the array of pixels, wherein each data line is coupled to a respective column of pixels; and a plurality of gate lines that is coupled to the array of pixels, wherein each transistor is interposed in a first direction between the respective pixel electrode and a given gate line in the plurality of gate lines, wherein a given pixel electrode is interposed in a second direction between two of the transistors, wherein adjacent pixel electrodes are non-overlapping in the second direction, and wherein the first and second directions are orthogonal.

16. The display defined in claim 15 wherein a first pixel, a second pixel, and a third pixel are in a given row, wherein the first pixel has a first pixel electrode and a first transistor, the second pixel has a second pixel electrode and a second transistor, and the third pixel has a third pixel electrode and a third transistor, and wherein the second transistor is interposed between the first and third pixel electrodes and the second pixel electrode is interposed between the first and third transistors.

17. The display defined in claim 15 wherein pixel electrodes in a given row and adjacent columns are separated by a non-zero distance in the first direction.

18. The display defined in claim 15 wherein the pixel electrodes are arranged in a zigzag layout.

19. The display defined in claim 15, further comprising: an opaque masking layer; and a plurality of pixel apertures in the opaque masking layer that each overlap a respective pixel electrode, wherein each pixel aperture has a first height that extends in the first direction, wherein the opaque masking layer between adjacent pixel apertures in the same column and different rows has a second height that extends in the first direction, and wherein the second height is greater than or equal to the first height.

20. The display defined in claim 15 wherein a given gate line of the plurality of gate lines is coupled to alternating pixels in first and second adjacent rows.
